

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
	11.0080 H hydrogen 2.2 -1 +1 1s¹																	4.0026 He helium — 2372.3 — 1s²	
1																			
2	6.94 Li lithium 0.98 +1 [He]2s¹	9.0122 Be beryllium 1.57 +2 [He]2s²																	
3	22.990 Na sodium 0.93 +1 [Ne]3s¹	24.305 Mg magnesium 1.31 +2 [Ne]3s²																	
4	39.098 K potassium 0.82 +1 [Ar]4s¹	40.078 Ca calcium 1 +2 [Ar]4s²	44.956 Sc scandium 1.36 +3 [Ar]3d¹4s²	47.867 Ti titanium 1.54 +2 +3 +4 [Ar]3d²4s²	50.942 V vanadium 1.63 +2 +3 +4 +5 [Ar]3d³4s²	51.996 Cr chromium 1.66 +2 +3 +6 [Ar]3d⁵4s¹	54.938 Mn manganese 1.55 +2 +3 +4 +6 +7 [Ar]3d⁵4s²	55.845 Fe iron 1.83 +2 +3 +6 [Ar]3d⁶4s²	58.933 Co cobalt 1.88 +2 +3 [Ar]3d⁷4s²	58.693 Ni nickel 1.91 +2 +3 [Ar]3d⁸4s²	63.546 Cu copper 1.9 +1 +2 [Ar]3d¹⁰4s¹	65.38 Zn zinc 1.65 +2 [Ar]3d¹⁰4s²	69.723 Ga gallium 1.81 +3 [Ar]3d¹⁰4s²4p¹	72.630 Ge germanium 2.01 -4 +2 +4 [Ar]3d¹⁰4s²4p²	74.922 As arsenic 2.18 -3 +3 +5 [Ar]3d¹⁰4s²4p³	78.971 Se selenium 2.55 -2 +2 +4 +6 [Ar]3d¹⁰4s²4p⁴	79.904 Br bromine 2.96 -1 +1 +3 +5 +7 [Ar]3d¹⁰4s²4p⁵	83.798 Kr krypton — +2 [Ar]3d¹⁰4s²4p⁶	
5	85.468 Rb rubidium 0.82 +1 [Kr]5s¹	87.62 Sr strontium 0.95 +2 [Kr]5s²	88.906 Y yttrium 1.22 +3 [Kr]4d¹5s²	91.222 Zr zirconium 1.33 +4 [Kr]4d²5s²	92.906 Nb niobium 1.6 +3 +5 [Kr]4d⁴5s¹	95.95 Mo molybdenum 2.16 +4 +6 [Kr]4d⁵5s¹	97 Tc technetium 2.1 +4 +7 [Kr]4d⁵5s²	101.07 Ru ruthenium 2.2 +2 +3 +4 +8 [Kr]4d⁷5s¹	102.91 Rh rhodium 2.28 +1 +3 [Kr]4d⁸5s¹	106.42 Pd palladium 2.2 +2 +4 [Kr]4d¹⁰	107.87 Ag silver 1.93 +1 [Kr]4d¹⁰5s¹	112.41 Cd cadmium 1.69 +2 [Kr]4d¹⁰5s²	114.82 In indium 1.78 +1 +3 [Kr]4d¹⁰5s²5p¹	118.71 Sn tin 1.96 -4 +2 +4 [Kr]4d¹⁰5s²5p²	121.76 Sb antimony 2.05 -3 +3 +5 [Kr]4d¹⁰5s²5p³	127.60 Te tellurium 2.1 -2 +2 +4 +6 [Kr]4d¹⁰5s²5p⁴	126.90 I iodine 2.66 -1 +1 +3 +5 +7 [Kr]4d¹⁰5s²5p⁵	131.29 Xe xenon 2.6 +2 +4 +6 +8 [Kr]4d¹⁰5s²5p⁶	
6	132.91 Cs caesium 0.79 +1 [Xe]6s¹	137.33 Ba barium 0.89 +2 [Xe]6s²	57–71	178.49 Hf hafnium 1.3 +4 [Xe]4f¹⁴5d²6s²	180.95 Ta tantalum 1.5 +5 [Xe]4f¹⁴5d³6s²	183.84 W tungsten 1.7 +4 +6 [Xe]4f¹⁴5d⁴6s²	186.21 Re rhenium 1.9 +4 +7 [Xe]4f¹⁴5d⁵6s²	190.23 Os osmium 2.2 +2 +3 +4 +8 [Xe]4f¹⁴5d⁶6s²	192.22 Ir iridium 2.2 +3 +4 [Xe]4f¹⁴5d⁷6s²	195.08 Pt platinum 2.2 +2 +4 [Xe]4f¹⁴5d⁹6s¹	196.97 Au gold 2.4 +1 +3 [Xe]4f¹⁴5d¹⁰6s¹	200.59 Hg mercury 1.9 +1 +2 [Xe]4f¹⁴5d¹⁰6s²	204.38 Tl thallium 1.8 +1 +3 [Xe]4f¹⁴5d¹⁰6s²6p¹	207.2 Pb lead 1.8 +2 +4 [Xe]4f¹⁴5d¹⁰6s²6p²	208.98 Bi bismuth 1.9 +3 +5 [Xe]4f¹⁴5d¹⁰6s²6p³	[209] Po polonium 2 -2 +2 +4 [Xe]4f¹⁴5d¹⁰6s²6p⁴	[210] At astatine 2.2 -1 +1 [Xe]4f¹⁴5d¹⁰6s²6p⁵	[222] Rn radon — — [Xe]4f¹⁴5d¹⁰6s²6p⁶	
7	[223] Fr francium 0.7 +1 [Rn]7s¹	[226] Ra radium 0.9 +2 [Rn]7s²	89–103	[267] Rf rutherfordium — +4 [Rn]5f¹⁴6d²7s²	[268] Db dubnium — — [Rn]5f¹⁴6d³7s²	[269] Sg seaborgium — — [Rn]5f¹⁴6d⁴7s²	[270] Bh bohrium — — [Rn]5f¹⁴6d⁵7s²	[269] Hs hassium — — [Rn]5f¹⁴6d⁶7s²	[277] Mt meitnerium — — [Rn]5f¹⁴6d⁷7s²	[281] Ds darmstadtium — — [Rn]5f¹⁴6d⁸7s²	[282] Rg roentgenium — — [Rn]5f¹⁴6d⁹7s²	[285] Cn copernicium — — [Rn]5f¹⁴6d¹⁰7s²	[286] Nh nihonium — — [Rn]5f¹⁴6d¹⁰7s²7p¹	[290] Fl flerovium — — [Rn]5f¹⁴6d¹⁰7s²7p²	[290] Mc moscovium — — [Rn]5f¹⁴6d¹⁰7s²7p³	[293] Lv livermorium — — [Rn]5f¹⁴6d¹⁰7s²7p⁴	[294] Ts tennessine — — [Rn]5f¹⁴6d¹⁰7s²7p⁵	[294] Og oganeson — — [Rn]5f¹⁴6d¹⁰7s²7p⁶	

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Periodic table of the elements

Sources: CIAAW 2024 (atomic masses) · NIST ASD (ionisation energies) · CRC Handbook (Pauling electronegativity)
Mendeleev 1.2.0 (electron configurations; reviewed overrides) · Oxidation states: selection adapted from Mendeleev 1.2.0

Atomic number

92

238.03

U

uranium

1.7

597.6

+3 +4 +5 +6

[Rn]5f³6d¹7s²

Radioactive sign

⚡

Atomic mass ([] = mass number of longest-lived isotope)

Symbol

Name

First ionisation energy (kJ/mol)

Electron configuration (ground state; predicted for Rf–Og)

	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71
Lanthanides	138.91 La lanthanum 1.1 +3 [Xe]5d¹6s²	140.12 Ce cerium 1.12 +3 +4 [Xe]4f¹5d¹6s²	140.91 Pr praseodymium 1.13 +3 +4 [Xe]4f²6s²	144.24 Nd neodymium 1.14 +3 [Xe]4f³6s²	[145] Pm promethium — +3 [Xe]4f⁴6s²	150.36 Sm samarium 1.17 +2 +3 [Xe]4f⁵6s²	151.96 Eu euporium — +2 +3 [Xe]4f⁶6s²	157.25 Gd gadolinium 1.2 +3 [Xe]4f⁷5d¹6s²	158.93 Tb terbium — +3 +4 [Xe]4f⁸6s²	162.50 Dy dysprosium 1.22 +3 [Xe]4f⁹6s²	164.93 Ho holmium 1.23 +3 [Xe]4f¹⁰6s²	167.26 Er erbium 1.24 +3 [Xe]4f¹¹6s²	168.93 Tm thulium 1.25 +2 +3 [Xe]4f¹²6s²	173.05 Yb ytterbium — +2 +3 [Xe]4f¹³6s²	174.97 Lu lutetium 1 +3 [Xe]4f¹⁴5d¹6s²
Actinides	[227] Ac actinium 1.1 +3 [Rn]6d¹7s²	232.04 Th thorium 1.3 +4 [Rn]6d²7s²	231.04 Pa protactinium 1.5 +4 +5 [Rn]5f²6d¹7s²	238.03 U uranium 1.7 +3 +4 +5 +6 [Rn]5f³6d¹7s²	[237] Np neptunium 1.3 +3 +4 +5 +6 +7 [Rn]5f⁴6d¹7s²	[244] Pu plutonium 1.3 +3 +4 +5 +6 [Rn]5f⁶7s²	[243] Am americium — +3 [Rn]5f⁷7s²	[247] Cm curium — +3 [Rn]5f⁷6d¹7s²	[247] Bk berkelium — +3 +4 [Rn]5f⁹7s²	[251] Cf californium — +3 [Rn]5f¹⁰7s²	[252] Es einsteinium — +3 [Rn]5f¹¹7s²	[257] Fm fermium — +3 [Rn]5f¹²7s²	[258] Md mendelevium — +2 +3 [Rn]5f¹³7s²	[259] No nobelium — +2 +3 [Rn]5f¹⁴7s²	[266] Lr lawrencium — +3 [Rn]5f¹⁴7s²7p¹

- Alkali metals
- Alkaline-earth metals
- Transition metals
- Post-transition metals
- Metalloids
- Nonmetals
- Halogens
- Noble gases
- Lanthanides
- Actinides
- Chemistry insufficiently characterised
- Two classifications